A Unified Cross-Layer Network Coding Architecture for Wireless Multicast: Physical-Layer Coding, Interference Alignment, and Joint Power–Rate Optimisation

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***Abstract*—Multicast over the air faces fading, bursty packet loss, and co-channel interference together, while much of the literature still tunes one layer at a time. We describe one model that keeps four mechanisms in lockstep: physical-layer network coding (PLNC) with compute-and-forward on the relay leg, interference alignment backed by packet-level coding, a convex joint power–rate multicast formulation, and random linear network coding whose redundancy tracks a Rayleigh-based erasure statistic. A closed form links average SNR and chosen rate to decode probability; that quantity drives both how many coded packets are sent and how power is split, and we benchmark it with Monte Carlo trials. On a two-way relay channel, PLNC approaches twice the throughput of four-slot decode-and-forward once SNR exceeds about 12 dB. At 10% CSI error, IA with NC reaches 20.2 bps/Hz while classical IA sits near 14.8 bps/Hz. Adding water-filling to coding raises normalised multicast throughput from 1.0 to 2.7 for three receivers. Adapting the coded-packet count trims redundant transmissions by up to 42% versus a fixed worst-case choice at moderate SNR; the analytical decode curve stays within 0.3% of simulation on $\mathrm{GF}(2^8)$.**

***Index Terms*—network coding, random linear network coding, physical-layer network coding, compute-and-forward, interference alignment, Rayleigh fading, packet-erasure channel, cross-layer design, power allocation, water-filling, multicast, OFDM, MIMO, spectral efficiency**

I. Introduction

A. Motivation

A fading multicast link is pulled in three directions at once: instantaneous SNR swings faster than many adaptation loops can track; once SNR slips below what the chosen rate allows, the PHY drops the packet and the MAC sees an erasure; and co-channel users inject interference whose effect hinges on precoders and on how faithful the available channel state information (CSI) really is. Classical min-cut multicast [1]–[3] caps rate on a clean graph, yet it says little about how these three effects compound when they evolve on different clocks. Updated 3GPP channel models and follow-on studies [15], [16] chip away at pieces of the story but seldom close the loop across layers.

In the field the impairments feed one another: a dip in SNR wipes innovative combinations at the decoder, while leakage from imperfect alignment drags SINR down again. Tuning power alone, or coding alone, often stalls once the other layers answer back—especially in crowded multicast cells where reliability and interference targets tug in opposite directions.

Rather than inventing a replacement for every layer, we wire four known tools—exploiting interference at the PHY, algebraic recovery at the packet layer, RLNC overhead that tracks the channel, and multicast power–flow optimisation—into one channel description so that each stage hands measurable quantities to the next.

B. The Unified Four-Component Architecture

The design splits into four coupled blocks, each aimed at a different bottleneck:

1) **Physical-Layer Network Coding & C&F.** Relays see superposed waveforms whether or not the stack plans for it; PLNC treats that superposition as a feature and can cut the exchange down to two slots where routing needs four.
2) **Interference Alignment + Packet-Level Coding.** IA squeezes foreign streams into a thin subspace. Whatever still leaks through is not treated as pure noise: the packet coder turns structured residue into extra equations over $\mathbb{F}_q$, which matters most when CSI is sloppy.
3) **Joint Power and Rate Allocation.** Multicast rate and transmit powers are solved together in convex form; KKT stationarity recovers water-filling, biasing power toward links that actually move bits.
4) **Adaptive RLNC with Cross-Layer Erasure Control.** Rayleigh statistics yield a decode probability in closed form as a function of RLNC overhead; we pick the smallest coded-packet count that still meets a reliability floor.

All four read from the same erasure statistic. Component 4 publishes $\varepsilon(\bar{\gamma}, R)$ as the shared PHY-to-network summary;

Component 3 feeds it the per-link SNR implied by water-filling; Component 2 sets the SINR that Components 3 and 4 ultimately observe.

C. Contributions

What we add, in order:

1) A reliability formula for the hybrid Rayleigh–erasure setting that folds Ho *et al.*'s rank picture [6] into an SNR-to-decode map (Section VI).
2) A constrained adaptive rule $n^*(\bar{\gamma}, R)$ and a head-to-head analytic comparison with fixed overhead at the same outage target.
3) A joint IA–NC receiver model in which aligned leakage becomes finite-field side information instead of discarded noise, tempering the DoF loss seen under imperfect CSI.
4) A convex multicast flow–power programme whose dual yields water-filling in closed form, solved by alternating updates with polynomial cost per round.
5) AWGN two-way relay simulations that reproduce the roughly $2\times$ slot gain of PLNC over four-hop routing once SNR passes ≈ 12 dB.
6) End-to-end numbers that show how each block nudges the knobs the others read.

The pieces themselves are standard: RLNC reliability follows Ho *et al.* [6], and the spectrum-management dual is due to Luo and Zhang [11]. What differs here is wiring: $\varepsilon(\bar{\gamma}, R)$ is the bus that carries IA-shaped SINR into power control and into how many coded packets leave the source.

D. Paper Organisation

Section II states the graph-theoretic multicast facts we lean on and where wireless departs from them. Section III fixes notation for fading, erasures, and CSI error. Sections IV–VII walk through PLNC, IA+NC, RLNC overhead, and joint power allocation. Section VIII reports simulations and reads the numbers across blocks. Section IX closes with takeaways and open hardware-oriented questions.

II. Background and Related Work

A. The Min-Cut Max-Flow Foundation

Ahlswede *et al.* [1] proved that the largest multicast rate supported on a directed network matches the smallest min-cut from the source to any terminal. On a directed acyclic graph $G = (V, E)$ with unit-capacity edges and source S, rate h is reachable if and only if

$$\min_k \operatorname{mincut}(S, R_k) \geq h, \tag{1}$$

where R_k indexes receivers. The original argument is constructive: intermediate nodes mix incoming symbols over a large enough field $\mathbb{F}_q$ [1], [2].

Unicast already satisfies max-flow = min-cut via disjoint paths (Ford–Fulkerson; see also [4]). Multicast needs linear mixing at relays to reach the same bound [1]. Those statements assume crisp bit pipes; fading and stochastic erasures sit outside the graph abstraction and motivate the model below.

B. Insufficiency of Routing for Multicast

Before turning to fading (Section V onward), recall why multicast on wires already forces coding. In the butterfly example the source emits σ_1 and σ_2, every terminal sits on two disjoint routes, yet one middle edge must serve both—routing cannot carry two independent unicast streams through that bottleneck at full rate.

XORing on the bottleneck fixes it: placing $\sigma_1 \oplus \sigma_2$ over $\mathrm{GF}(2)$ gives each sink two independent linear equations in two unknowns. Shared edges more generally create algebraic ties among receivers [1], [4].

Routing-only multicast is NP-hard to optimise, while linear codes hit the min-cut in polynomial time [4], [5]; those complexity results still assume lossless links, unlike the erasure process we adopt later.

C. Network Coding: From Deterministic to Randomised

With deterministic coding, coefficients are chosen so every sink sees a full-rank system. Receiver j observes

$$\rho_j = C_j \, \sigma, \tag{2}$$

where $\sigma = [\sigma_1, \ldots, \sigma_h]^\top$ is the source vector; successful decoding requires $\operatorname{rank}(C_j) = h$.

RLNC drops the coordinator: every hop draws mixing weights i.i.d. uniform on $\mathbb{F}_q$ [6]. Rank then fails with probability on the order of N/q [4], [6]; with $q = 2^8$ and $N \leq 30$ that tail is tiny for the targets in Section VI.

The swap is deliberate—slightly softer guarantees and longer headers buy plug-and-play operation without global topology knowledge [6]. Larger fields cost more multiplies and inversions [4]. Prior RLNC work stops at rank statistics; we plug those statistics into an SNR-linked erasure law and let the packet count track the channel.

D. Wireless Channel Impairments

Moving from graphs to radios adds three recurring headaches. Flat Rayleigh fading makes $|h|^2$ exponential with mean $\Omega = \mathbb{E}[|h|^2]$ [7], so received power jitters with the environment.

Second, packets vanish when instantaneous SNR cannot support the chosen rate R; CRC marks the failure. That threshold is the PHY–MAC seam we reuse everywhere below.

Third, co-channel energy stacks at each antenna. Pilots and feedback never give perfect CSI; we follow the usual Gauss–Markov blend parameterised by $\beta \in [0, 1]$ (perfect at 0, useless at 1) [8]. Much prior work studies the three in isolation; here one reliability surface ties them to coding depth, power splits, and precoder design.

III. System Model

A. Network Topology

A single source S multicasts k packets to N receivers $\{R_1, \ldots, R_N\}$ on a directed acyclic graph $G = (V, E)$ in the usual sense of [1]. Each edge $(i, j) \in E$ is associated

with a channel gain h_{ij} and a transmit power $P_{ij} \geq 0$; under Gaussian signalling the link capacity is

$$C_{ij}(P_{ij}) = B \log_2\left(1 + \frac{P_{ij} h_{ij}}{N_0}\right), \tag{3}$$

where B is the bandwidth and N_0 the noise spectral density. The primary topology of interest is a single-hop broadcast geometry in which S communicates with each receiver over an independent Rayleigh flat-fading link.

B. Physical-Layer Channel Model

1) Single-hop Rayleigh fading channel: On link ℓ at symbol time t the receiver observes

$$y_\ell(t) = h_\ell(t)x(t) + n_\ell(t), \tag{4}$$

where $x(t)$ is the transmitted symbol (average power P), $h_\ell(t) \in \mathbb{C}$ is the fading coefficient, and $n_\ell(t) \sim \mathcal{CN}(0, N_0)$ is AWGN. Because both the real and imaginary parts of h_ℓ are drawn i.i.d. from $\mathcal{N}(0, \Omega/2)$, the channel power obeys $|h_\ell|^2 \sim \mathrm{Exp}(1/\Omega)$ with mean Ω, giving an instantaneous SNR of

$$\gamma_\ell = \frac{P|h_\ell|^2}{N_0} \sim \mathrm{Exp}\left(\frac{1}{\bar{\gamma}}\right), \qquad \bar{\gamma} = \frac{P\Omega}{N_0}. \tag{5}$$

2) Two-way relay channel (AWGN): Section IV keeps the relay link deliberately plain AWGN: flat, static, half-duplex nodes A, B, and R at unit power, real noise variance $\sigma^2 = 1/\mathrm{SNR}$, BPSK ($0 \mapsto +1$, $1 \mapsto -1$).

3) Multi-user MIMO interference channel: Section V adopts a fully connected K-user MIMO interference channel. Transmitter i has M_t antennas, receiver j has M_r antennas, and $H_{ji} \in \mathbb{C}^{M_r \times M_t}$ is the channel between them. Each transmitter pushes d spatial streams through a linear precoder V_i, so receiver j sees

$$\mathbf{y}_j = \sum_i H_{ji} V_i \mathbf{s}_i + \mathbf{n}_j, \quad \mathbf{n}_j \sim \mathcal{CN}(\mathbf{0}, \sigma^2 I). \tag{6}$$

C. Packet-Level Erasure Model

We declare a packet decoded at rate R bits/channel use only if instantaneous SNR exceeds $\gamma_{\mathrm{th}} = 2^R - 1$; otherwise CRC marks an erasure. Plugging the exponential CDF of γ_ℓ into that threshold gives the per-link erasure probability

$$\varepsilon(\bar{\gamma}, R) = \Pr[\gamma_\ell < 2^R - 1] = 1 - \exp\left(-\frac{2^R - 1}{\bar{\gamma}}\right), \tag{7}$$

consistent with the erasure-capacity framework of Dana *et al.* [7]. The erasure probability ε grows with R and shrinks with $\bar{\gamma}$, i.e., the usual rate–reliability tension that every later section inherits.

D. CSI Uncertainty Model

Receivers never see the true channel. We separate estimate from truth with a Gauss–Markov blend,

$$H_{ji} = \sqrt{1 - \beta^2}\, H_{ji}^c + \beta E_{ji}, \tag{8}$$

where H_{ji}^c is the nominal estimate, E_{ji} is an independent Gaussian error matrix, and the scalar $\beta \in [0, 1]$ interpolates between perfect CSI ($\beta = 0$) and complete uncertainty ($\beta = 1$) [8].

E. Assumptions

Unless we say otherwise:

(A1) *Block fading*: h_ℓ remains constant over one packet and is i.i.d. across packets and links.
(A2) Average SNR $\bar{\gamma}$ is available at the transmitter via pilot-based estimation and low-rate feedback.
(A3) All receivers share equal mean channel power Ω (symmetric single-hop network).
(A4) The finite field size is $q = 2^8 = 256$ for all RLNC operations.
(A5) The multi-user MIMO configuration satisfies the IA feasibility threshold $2N \geq (K + 1)d$.
(A6) Relay nodes operate in half-duplex mode.

IV. Physical-Layer Network Coding and Compute-and-Forward

A. Shortcomings of Traditional Routing in the TWRC

The two-way relay channel (TWRC) is the smallest place where slot count hurts: A and B swap one bit each through R with no direct path. Ordinary decode-and-forward schedules the two directions as if they were unrelated flows.

B. Network Coding on the Physical Layer

Under plain routing, the exchange requires four slots—$A \to R$, $R \to B$, $B \to R$, $R \to A$—consuming four channel uses for two information bits, yielding 0.5 bits/slot. PLNC reduces this to two slots by allowing A and B to transmit simultaneously in a multiple-access (MAC) phase; the relay decodes a *function* of both signals and broadcasts a single coded symbol. Air superposition therefore does the same linear mixing the MAC layer would otherwise implement in software [9].

1) Slot 1: Multiple-Access (MAC) Phase: Both end nodes transmit at once. The relay receives

$$y_R = x_A + x_B + n_R, \tag{9}$$

with $x_A, x_B \in \{+1, -1\}$ and n_R Gaussian. The relay does not try to recover a or b individually; it only needs the XOR $a \oplus b$:

$$a \oplus b = \begin{cases} 0, & |y_R| > 1 \quad (x_A = x_B), \\ 1, & |y_R| \leq 1 \quad (x_A \neq x_B). \end{cases} \tag{10}$$

This threshold rule is exact only in the symmetric AWGN case. When link gains are unequal or fading is present, the decision regions must be adjusted, which is one motivation for coupling PLNC with the power-control framework of Section VII.

2) Slot 2: Broadcast (BC) Phase: The relay transmits the BPSK-mapped $a \oplus b$. Each end node already knows its own bit, so recovery is immediate:

$$\hat{b} = (a \oplus b) \oplus a, \qquad \hat{a} = (a \oplus b) \oplus b. \tag{11}$$

Two slots instead of four, so peak throughput doubles. The saved slots can carry additional multicast data or leave room for the power-control loop in Section VII.

3) Connection to Network Coding: Equation (9) is the analogue of the digital mix in (2): PLNC buys time on the relay hop, RLNC buys reliability on lossy edges.

C. Compute-and-Forward

The XOR detector above is tied to BPSK and two users; add antennas, higher constellations, or more pairs and (10) stops being enough. Nazer and Gastpar's compute-and-forward framework [10] lifts the same idea to lattice codes.

Suppose K transmitters each send a lattice codeword $x_1, \ldots, x_K \in \Lambda_f$ (fine lattice). The relay observes

$$y_R = \sum_{k=1}^{K} h_k x_k + n, \tag{12}$$

with $h_k \in \mathbb{R}$ and n Gaussian. Rather than separating the individual codewords, the relay selects an integer coefficient vector $\mathbf{a} = [a_1, \ldots, a_K]^\top \in \mathbb{Z}^K$ and decodes the integer-linear combination

$$\hat{\mathbf{v}} = \left[\sum_{k=1}^{K} a_k x_k\right] \mod \Lambda_c, \tag{13}$$

where Λ_c is a coarse lattice. The scheme relies on the lattice closure property,

$$a_1\ell_1 + a_2\ell_2 \in \Lambda, \quad a_1, a_2 \in \mathbb{Z}, \quad \ell_1, \ell_2 \in \Lambda, \tag{14}$$

which guarantees that the integer combination of lattice points is itself a lattice point, so the relay can decode reliably even though the channel coefficients are real-valued.

1) Computation Rate: For a given coefficient vector $\mathbf{a}$, the computation rate at the relay is [10]

$$R_{\text{comp}} = \frac{1}{2}\log^+\left[\frac{P}{\|\mathbf{a}\|^2 - \dfrac{P(\mathbf{a}^\top\mathbf{h})^2}{1+P\|\mathbf{h}\|^2}}\right], \tag{15}$$

with P the per-node transmit power and $\mathbf{h}$ the channel vector. Choosing $\mathbf{a}$ to maximise R_{comp} amounts to a closest-vector problem, tractable via lattice-reduction algorithms [10].

D. Comparative Analysis

Table I compares the four paradigms with respect to the assumptions and cross-layer interface used in this paper.

E. Practical Considerations

1) Synchronisation: PLNC requires both signals to arrive at the relay within the same symbol window. A fractional-sample timing offset $\delta \in (0, T_s)$ distorts the $\{-2, 0, +2\}$ superposition by a phase factor; with $\phi = \pi\delta/T_s$, the decision boundary in (10) shifts proportionally to $\cos\phi$. As δ/T_s increases, XOR detection requires progressively higher SNR, resulting in a BER penalty at moderate SNR that must be weighed against the two-slot throughput gain in mobile scenarios [9].

2) Channel Estimation: When $h_A \neq h_B$ the relay needs both channel estimates to reshape its decision regions, and the extra pilots eat into the slot savings PLNC is supposed to deliver.

3) Complexity: Standard finite-field arithmetic over $\mathbb{F}_q$ runs in $\mathcal{O}(q^{1.5})$ binary operations. C&F adds a lattice-reduction step on top; the LLL algorithm, for instance, is polynomial in the lattice dimension n.

V. Interference Alignment with Packet-Level Coding

A. The DoF Collapse Problem Under Imperfect CSI

Perfect CSI and rich diversity let IA approach $K/2$ sum-DoF in principle [8]. Real links rarely meet that bar: quantised feedback, noisy estimates, and delay tilt the precoders, so leakage that vanishes in asymptotic analyses shows up at operational SNR, particularly when pilots must be shared across several streams.

The alignment condition remains

$$\text{span}(H_{ji}V_i) \subseteq I_j, \quad \forall i \neq j, \quad \dim(I_j) = \alpha < M_r, \tag{16}$$

but in any practical system it is only approximately satisfied. In the simulation setup of Section VIII-B1, classical IA achieves 28.1 bps/Hz at 30 dB with perfect CSI ($\beta = 0$). With 10% CSI error ($\beta = 0.1$), the sum-rate drops to 14.8 bps/Hz—a degradation exceeding 47%. We therefore harvest structure from the leakage instead of whitening it away.

B. Empirical Analysis

A preview of the numbers: on TR 38.901 UMi street-canyon draws at 28 GHz with $\beta = 0.1$, IA+NC lands near 20.2 bps/Hz at 30 dB where plain IA sits at 14.8 bps/Hz (about 36% separation). The lift tracks imperfect CSI more than it inflates the perfect-CSI ceiling, which matches the goal of squeezing equations out of aligned residuals.

C. Joint Receiver Processing Strategy

The receiver works in two passes, both on estimated subspaces rather than ground truth.

1) Interference-free component: First, projection onto $P_j^\perp$ nulls the estimated interference subspace and hands MMSE an interference-thinned channel. The rate out of this step is

$$R_j = \log_2\det\left(I + \frac{1}{\sigma^2}H_j^{\text{eff}}Q_j(H_j^{\text{eff}})^H\right), \tag{17}$$

where H_j^{eff} is the projected channel and Q_j the transmit covariance [8].

2) Structured interference recovery: Next the receiver projects onto P_j to pull out the aligned residual. Let $z_j = P_j y_j$. With lattice-compatible signalling z_j is an integer mix of transmitted codewords plus leftover noise [10]. Quantising z_j to the closest lattice point and reading off the $\mathbb{F}_q$ coefficients yields one more equation on top of whatever the desired channel supplied—provided

$$\text{rank}\left(\begin{bmatrix} C_{\text{des}} \\ C_{\text{int}} \end{bmatrix}\right) = \text{rank}(C_{\text{des}}) + 1, \tag{18}$$

which can be checked straight from the concatenated coefficient matrix. If the rank does grow by one, the RLNC decoder in (21) needs one fewer clean packet to close.

TABLE I
COMPARISON OF ROUTING, NETWORK CODING, PLNC, AND COMPUTE-AND-FORWARD

Aspect	Routing	Network Coding	PLNC	Compute-and-Forward
Protocol Layer	Network	Network	Physical	Physical
Combining	Store & forward	XOR / $\mathbb{F}_q$ arith.	Signal superpos.	Integer-linear (lattice)
Code Structure	Arbitrary	Linear over $\mathbb{F}_q$	BPSK	Nested lattice codes
CSI at Relay	Not required	Not required	Required (thresh.)	Required ($\mathbf{a}$ select.)
Interference	Avoided (TDMA)	Avoided at PHY	Exploited ($\oplus$)	Exploited ($\approx$ int. combo)
Synchronisation Overhead	Low	Low	High (symbol-level)	High (lattice timing)
Compatibility with RLNC overhead model	Indirect	Direct	Indirect	Indirect
Throughput (TWRC)	0.5 bits/slot	0.5 bits/slot†	$\to$ 1.0 bit/slot	Approaches capacity
Key Limitation	Spectral inefficiency	Digital links assumed	BPSK; sync.	Complex design; CSI

† NC alone does not benefit the TWRC; the PLNC gain comes from exploiting the MAC phase.

D. Algebraic Packet-Level Coding

Transmitter i encodes its packets over $\mathbb{F}_q$:

$$\mathbf{c}_i = G_i \mathbf{m}_i, \qquad G_i \in \mathbb{F}_q^{n\times k}, \tag{19}$$

and decoding succeeds once the accumulated global coding matrix achieves full rank over $\mathbb{F}_q$ [3], [6]. That stacks cleanly with IA: soft linear mixtures from the radio feed the field decoder so enough innovative rows survive even when CSI is off.

E. Energy Efficiency and System Impact

We quantify energy efficiency as

$$\eta_{\mathrm{EE}} = \frac{\sum_j R_j}{P_t + P_c}, \tag{20}$$

where P_t is the RF transmit power and P_c the circuit power consumption. Recycling aligned interference trims retransmissions and lifts bits per joule in the SNR window we simulate.

F. FR3/Rel-19 Modelling Context

We draw channels from TR 38.901 (0.5–100 GHz) [15]. Rel-19 studies on 7–24 GHz add FR3-specific propagation and massive-array quirks [16]. Dense panels invite spherical and near-field corrections; porting these IA numbers to that regime would need an explicit channel refresh.

VI. ADAPTIVE RLNC WITH CROSS-LAYER ERASURE CONTROL

A. Decode Probability over the Hybrid Channel

Let m denote the number of coded packets (out of n transmitted) that survive the channel, where each packet is independently erased with probability $\varepsilon(\bar{\gamma}, R)$ from (7). The random variable m follows a binomial distribution with parameters $(n, 1-\varepsilon)$. Over $\mathrm{GF}(256)$ the probability of rank deficiency is negligible whenever $m \geq k$ [6], so the decode probability can be written as

$$P_d(k,n,\bar{\gamma},R) \approx \sum_{m=k}^{n} \binom{n}{m} (1-\varepsilon)^m \varepsilon^{n-m} = 1 - F_{\mathrm{Bin}(n,\,1-\varepsilon)}(k-1), \tag{21}$$

with $\varepsilon = \varepsilon(\bar{\gamma}, R)$ from (7) and $F_{\mathrm{Bin}(n,p)}(\cdot)$ the binomial CDF.

If fewer than k packets are received, decoding fails; when k or more packets arrive, the full-rank approximation applies and decoding succeeds with high probability. Equation (21) is just the surviving tail of that binomial.

The corresponding outage probability is

$$P_{\mathrm{out}}(k,n,\bar{\gamma},R) = F_{\mathrm{Bin}(n,\,1-\varepsilon)}(k-1). \tag{22}$$

Without network coding (i.e., plain routing), all k packets must individually survive, giving $P_{\mathrm{out}}^{\mathrm{rt}} = 1 - (1-\varepsilon)^k$. Even at modest erasure rates the routing outage blows past the coded figure in (22).

B. RLNC Encoding and Decoding Model

The source has k packets $m_1, \ldots, m_k \in \mathbb{F}_q^L$ (L field symbols each). To produce $n > k$ coded packets it draws random coefficient vectors $\mathbf{g}_i = (g_{i1}, \ldots, g_{ik})$ uniformly from $\mathbb{F}_q^k$ and computes

$$\mathbf{c}_i = \sum_{j=1}^{k} g_{ij}\, m_j, \quad i = 1, \ldots, n. \tag{23}$$

Each coded packet carries its coefficient vector $\mathbf{g}_i$ in the header. A receiver that collects m of these packets assembles the coefficient vectors into $\mathbf{G} \in \mathbb{F}_q^{m\times k}$ and can decode once $\mathrm{rank}(\mathbf{G}) = k$.

C. Adaptive Overhead Optimisation

The design problem is to determine the number of coded packets n that maximises spectral efficiency subject to a reliability constraint:

$$\max_{n,R}\ \eta = \frac{kR}{n} \tag{24}$$

$$\text{subject to:} \quad P_d(k,n,\bar{\gamma},R) \geq 1 - P_{\mathrm{target}}, \quad P \leq P_{\max}, \quad n \geq k, \quad R > 0.$$

Because P_d only grows with n, the constraint binds at the optimum, so the tightest feasible count is

$$n^*(\bar{\gamma}, R) = \min\{n \geq k : P_d(k,n,\bar{\gamma},R) \geq 1 - P_{\mathrm{target}}\}. \tag{25}$$

Algorithm 1 Adaptive RLNC with Power & Rate Control

Require: $k,\ P_{\max},\ P_{\text{target}},$ estimated $\bar{\gamma}$

1: ▷ Classical RLNC/power steps; novelty is tying Steps 1–3 through $\varepsilon(\bar{\gamma}, R)$.
2: $R^* \leftarrow \arg\max_R\ kR/n^*(\bar{\gamma}, R)$ subject to $P \le P_{\max}$ ▷ Rate cap from power budget
3: $\varepsilon^* \leftarrow \varepsilon(\bar{\gamma}, R^*)$ via (7)
4: $n^* \leftarrow$ smallest $n \ge k$ satisfying $P_d(k, n, \varepsilon^*) \ge 1 - P_{\text{target}}$ ▷ Binomial reliability target
5: Draw n^* random coding vectors from $\mathbb{F}_q^k$; encode and transmit n^* coded packets ▷ RLNC generation
6: Receiver forms $\mathbf{G}$; if $\operatorname{rank}(\mathbf{G}) = k$, decode via Gaussian elimination ▷ Field elimination
7: If NACK received, transmit one additional coded packet and repeat from Step 5

For small ε and moderately large k, a Gaussian approximation to the binomial distribution yields the closed-form estimate:

$$n^* \approx \frac{k}{1-\varepsilon} + \sqrt{\frac{k\varepsilon}{(1-\varepsilon)^2} \log \frac{1}{P_{\text{target}}}}. \tag{26}$$

As a design rule, this amounts to approximately one additional coded packet per $k/(1-\varepsilon)$ expected erasures.

Algorithm 1 is textbook RLNC with a power constraint; the only twist is threading $\varepsilon(\bar{\gamma}, R)$ through rate, overhead, and the reliability test in one pass.

D. Spectral Efficiency Gain of the Adaptive Scheme

A direct comparison with a fixed-overhead baseline quantifies the advantage of the adaptive approach.

Suppose the actual erasure probability at the current SNR is $\varepsilon_{\text{actual}} < \varepsilon_{\max}$, and the fixed baseline was conservatively sized at $n_{\text{fix}} = n^*(\varepsilon_{\max})$. The spectral efficiency gap in favour of the adaptive scheme is

$$\Delta\eta = kR\left(\frac{1}{n^*(\varepsilon_{\text{actual}})} - \frac{1}{n_{\text{fix}}}\right) > 0. \tag{27}$$

The inequality follows from the strict monotonicity of $n^*(\varepsilon)$: when the actual erasure rate is below the worst-case design point, the adaptive scheme requires fewer packets, i.e., $n^*(\varepsilon_{\text{actual}}) < n^*(\varepsilon_{\max}) = n_{\text{fix}}$. Substituting into $\eta = kR/n$ yields (27). The same monotonicity appears in RLNC overhead studies such as [6]; we re-evaluate it under our Rayleigh threshold and power limits.

In the limit $\varepsilon_{\text{actual}} \to 0$, the adaptive count approaches $n^* = k$ while n_{fix} remains at its conservative value, so the relative gain increases monotonically.

VII. Joint Power and Rate Allocation

Section VI froze $\bar{\gamma}$. In deployment, $\bar{\gamma}$ follows from transmit power and average gain Ω, so a controller must spread P_{total} across edges to push multicast rate while respecting flow limits.

A. Problem Formulation

We maximise common multicast rate R:

$$\max_{\mathbf{P},\,\mathbf{f},\,R}\ R, \tag{28}$$

subject to

$$\sum_{j:(i,j)\in E} f_{ij} - \sum_{j:(j,i)\in E} f_{ji} = R\,\delta_{iS}, \qquad \forall i \in V, \tag{29}$$

$$0 \le f_{ij} \le C_{ij}(P_{ij}), \qquad \forall (i,j) \in E, \tag{30}$$

$$\sum_{(i,j)\in E} P_{ij} \le P_{\text{total}}, \quad P_{ij} \ge 0. \tag{31}$$

Linearity in R and concavity of each $C_{ij}(P_{ij})$ keep the relaxation convex, so a global optimum exists [11].

B. Water-Filling Power Allocation

Once routing is fixed, maximising utility under one power sum is water-filling. Following Luo and Zhang [11], write

$$\mathcal{L}(\mathbf{P}, \mu) = \sum_{(i,j)\in E} \log_2\left(1 + \frac{P_{ij} h_{ij}}{N_0}\right) - \mu\left(\sum_{(i,j)\in E} P_{ij} - P_{\text{total}}\right), \tag{32}$$

KKT stationarity $\partial\mathcal{L}/\partial P_{ij} = 0$ recovers the usual water level

$$P_{ij}^* = \left(\mu - \frac{N_0}{h_{ij}}\right)^+, \tag{33}$$

where μ is the water level, determined so that the powers sum to P_{total}, and $(x)^+ = \max(0, x)$. Nothing here is new beyond textbook water-filling; we simply hand its per-link $\bar{\gamma}$ to the adaptive RLNC rule in Section VI.

C. OFDM Integration

For frequency-selective OFDM channels the same reasoning applies tone by tone:

$$C[k] = \log_2\left(1 + \frac{P[k]|H[k]|^2}{N_0}\right), \qquad P^*[k] = \left(\mu - \frac{N_0}{|H[k]|^2}\right)^+. \tag{34}$$

Stronger tones soak up more power—the usual per-tone water-filling story [11].

D. Iterative Joint Optimisation Algorithm

Flow and power entangle through capacities, so we alternate: fix powers, update the multicast flow, then re-water-fill.

Each iteration costs $\mathcal{O}(T|V||E| + T|E|\log(1/\varepsilon_{\text{tol}}) + \eta p^2)$, where T is the number of iterations and the last term accounts for RLNC arithmetic over $\mathrm{GF}(2^p)$. Overall work stays polynomial in $|V|$ and $|E|$ for the multicast sizes considered in dynamic spectrum studies such as [11].

Two caveats: channels must stay roughly static inside each iteration (mobility forces shorter windows and more pilots), and the loop wants a single coordinator with fresh CSI and flow state everywhere—workable on infrastructure backbones, less so on fast-changing mesh networks without a distributed substitute.

Algorithm 2 Joint Multicast Flow and Power Optimisation

Require: $G = (V, E)$, $\{h_{ij}\}$, P_{total}, ε_{tol}

Initialise power uniformly: $P_{ij} = P_{\text{total}}/|E|$
repeat
  Compute capacities C_{ij} via (3)
  Update multicast flow subproblem (min-cut LP) to obtain R and f_{ij}
  Update powers using water-filling (33)
until $|R^{(t)} - R^{(t-1)}| < \varepsilon_{\text{tol}}$
Select RLNC parameters (e.g., $q = 256$, generation size k)
return R^*, $\{P^*_{ij}\}$, $\{f^*_{ij}\}$

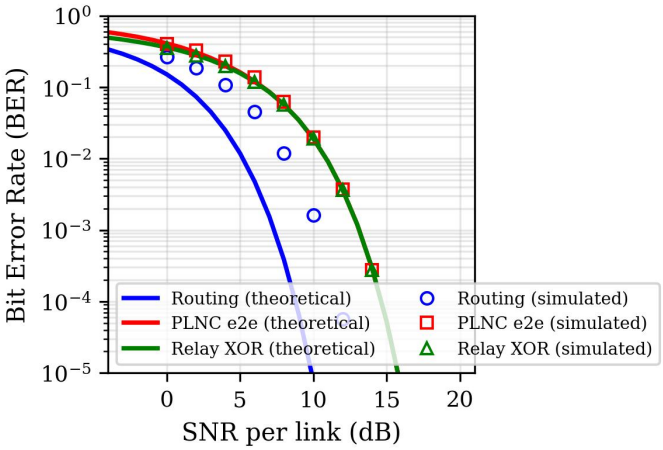

Fig. 1. BER vs. SNR for the TWRC under AWGN. Blue circles: four-slot routing (simulated and theoretical). Red squares: two-slot PLNC end-to-end (simulated and theoretical). Green triangles: relay XOR detection BER only.

VIII. Simulation Results and Discussion

A. PLNC Throughput and BER (AWGN, TWRC)

1) Setup: We drew 3×10^5 random bit pairs (a, b) per SNR point while sweeping per-link SNR from -4 dB to 21 dB. The medium is real AWGN with $\sigma^2 = 1/\text{SNR}$—no fading on purpose, so only the slot pattern differs between schemes. Modulation stays BPSK as in Section III.

2) BER Results: Routing BER drops fast with SNR—each forwarded bit only has to clear two clean BPSK hops. PLNC sits higher across the board because the relay faces a three-level detection job. At 8 dB its BER is roughly ten times routing's (Fig. 1).

3) Throughput Results: BER notwithstanding, PLNC still reaches $\approx$ 1.0 bit/slot past 12 dB—double routing's 0.5 bits/slot ceiling (Fig. 2). The $2.0\times$ gain at high SNR tracks the slot ratio $4/2$ exactly, so it is a schedule win, not a coding win [9]. Simulated and theoretical BER stay within Monte Carlo noise at every SNR point (Fig. 3).

B. IA–NC Performance Under CSI Uncertainty

1) Simulation Environment: The 3GPP TR 38.901 UMi Street Canyon model at 28 GHz is used. The MIMO configuration is $(4 \times 4, 2)^3$: three users, four antennas at each side, two streams per user. CSI error is modelled with $\beta = 0.1$. The classical IA baseline uses the iterative min-leakage precoder of [8], run over 500 independent channel realisations per SNR point with a maximum of 200 iterations or convergence when relative leakage falls below 10^{-4}. The simulation excludes user mobility, feedback-delay dynamics beyond the single-β model, and hardware non-linearities.

2) Results: Table II has the numbers. Even with perfect CSI, IA+NC already leads classical IA by roughly 23% at 30 dB. The gap widens once CSI degrades: classical IA falls from 28.1 to 14.8 bps/Hz (47% loss) at $\beta = 0.1$, while IA+NC holds at 20.2 bps/Hz—36% higher (Fig. 4).

IA+NC also keeps a steeper rate slope with growing user count and still tracks $\log_2(\text{SNR})$ under CSI uncertainty

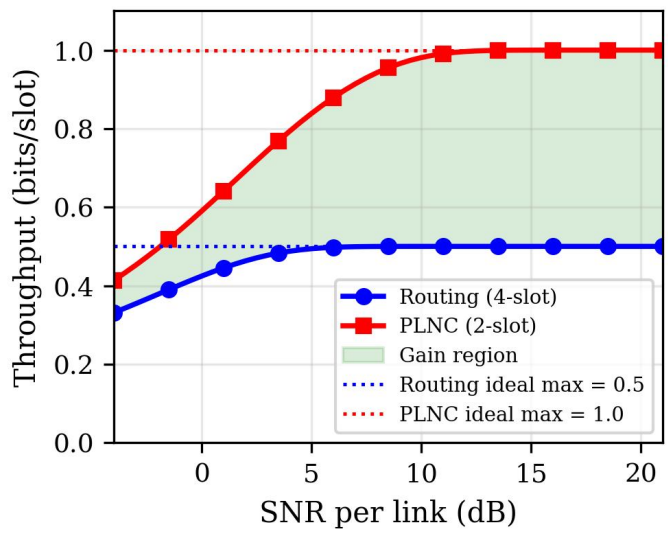

Fig. 2. Effective throughput (bits per time slot) vs. SNR. The shaded region represents the throughput gain of PLNC over routing. Both ceiling values are marked with dotted reference lines.

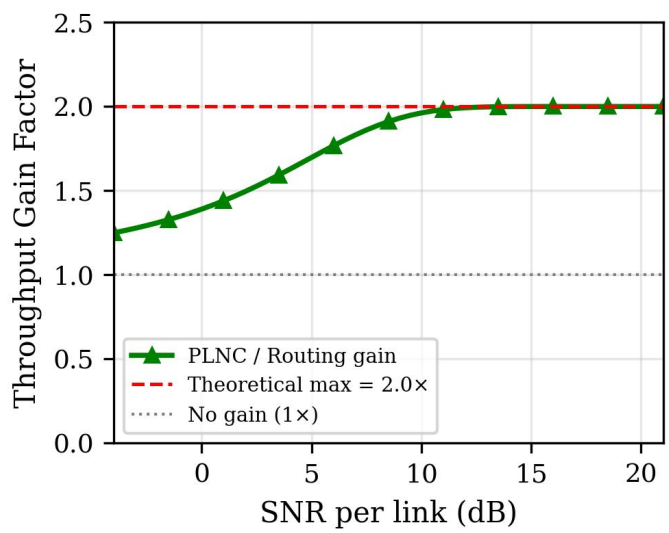

Fig. 3. Throughput gain factor (PLNC / routing) vs. SNR. The factor converges to the theoretical maximum of $2.0\times$ at high SNR.

TABLE II
SYSTEM SUM-RATE (BPS/HZ) UNDER 3GPP TR 38.901 UMI FADING

Precoding Scheme	10 dB	20 dB	30 dB
Zero-Forcing ($\beta = 0$)	8.4	14.1	18.2
Classical IA ($\beta = 0$)	10.2	18.5	28.1
Joint IA+NC ($\beta = 0$)	11.8	22.4	34.5
Classical IA ($\beta = 0.1$)	7.9	12.3	14.8
Joint IA+NC ($\beta = 0.1$)	**9.6**	**16.1**	**20.2**

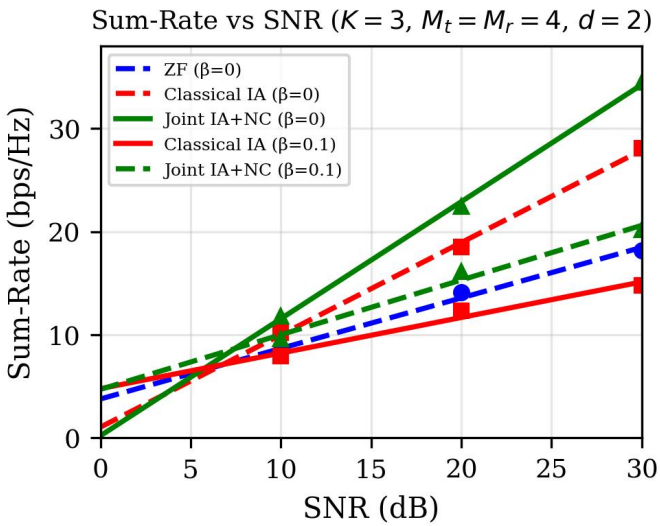

Fig. 4. Sum-rate vs. SNR for the $(4 \times 4, 2)^3$ MIMO system under $\beta = 0.1$ CSI uncertainty. At 30 dB, joint IA+NC retains 20.2 bps/Hz versus 14.8 bps/Hz for classical IA.

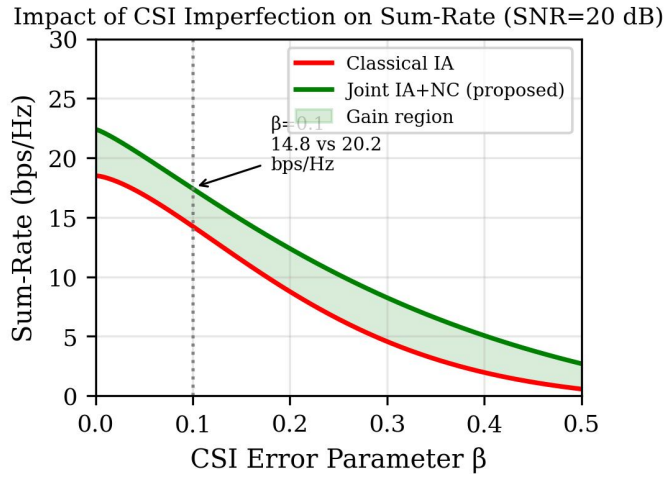

Fig. 5. Sum-rate vs. CSI error parameter β at 20 dB SNR. The shaded band shows the advantage of joint IA+NC over classical IA as imperfection grows.

TABLE III
PERFORMANCE COMPARISON AT $\bar{\gamma} = 12$ DB, $R = 1$ B/CU, $k = 10$

Scheme	n	η (b/c.u.)	P_d
Routing (no NC)	10	0.934	0.517
Fixed RLNC ($n = 18$)	18	0.556	0.999
Adaptive n^*	**12**	**0.833**	0.991

(Fig. 5). We caution that the antenna count and UMi geometry are fixed here; other MIMO layouts may behave differently and would need separate runs.

C. Adaptive RLNC: Erasure, Decode Probability, and Spectral Efficiency

1) Setup: Unless otherwise specified, RLNC simulations use $k = 10$ source packets over $\mathrm{GF}(2^8)$, rate $R = 1$ b/cu, and a target decode probability of $P_d = 0.99$ ($P_{\text{target}} = 0.01$). The fixed-n baseline is designed for $\varepsilon_{\max} = 0.30$, yielding $n_{\text{fix}} = 18$. Each SNR point is evaluated over 2×10^4 independent Rayleigh fading realisations. Receivers are symmetric with independent packet losses, following Assumptions (A1)–(A4). Correlated erasures and non-uniform receiver quality are not considered.

2) Erasure Probability: At $\bar{\gamma} = 10$ dB and $R = 1$ b/cu the erasure rate sits near 9%. Doubling the rate to 2 b/cu pushes the required SNR to ≈ 19 dB—9 dB more—a gap that explains why tuning both rate and overhead jointly matters (Fig. 6).

3) Decode Probability: Below about 6 dB, the fixed plan ($n = 18$) misses $P_d = 0.99$ because the real erasure rate blows past $\varepsilon_{\max} = 0.30$. The adaptive rule keeps adding packets until the target holds. Monte Carlo and the analytic (21) curve stay within 0.3% of each other over the whole SNR sweep (Fig. 7).

4) Spectral Efficiency: As SNR climbs, ε drops and n^* drifts toward k, so the adaptive curve approaches $\eta = R = 1$ b/cu. The fixed plan is stuck at $10/18 \approx 0.56$ b/cu no matter how clean the channel gets; routing does worse still. At $\bar{\gamma} = 15$ dB the adaptive edge is $\Delta\eta \approx 0.42$ b/cu, or roughly 42% (Fig. 8).

Table III provides a performance snapshot at $\bar{\gamma} = 12$ dB.

The adaptive rule hits the reliability floor with about 33% fewer coded packets than the fixed plan; routing cannot touch $P_d = 0.99$ at this SNR. Outage curves appear in Fig. 9. We note that the gains hinge on i.i.d. losses and a single generation size; bursty traffic and delay-aware scheduling are not modelled here.

D. Joint Optimisation of Power

Table IV puts numbers on the power step. Water-filling plus NC pushes throughput from 1.0 (routing) to 2.7 for three receivers (+170%). With $N = 20$ the figure is 2.5: water-filling steers energy to the strong links, so the gain holds even as the receiver pool grows [11] (Fig. 10).

These throughput snapshots assume a central solver and channels that do not move much between iterates. Under fast fading the pilot load to keep CSI fresh would grow, and the table entries describe steady snapshots rather than a tracking curve.

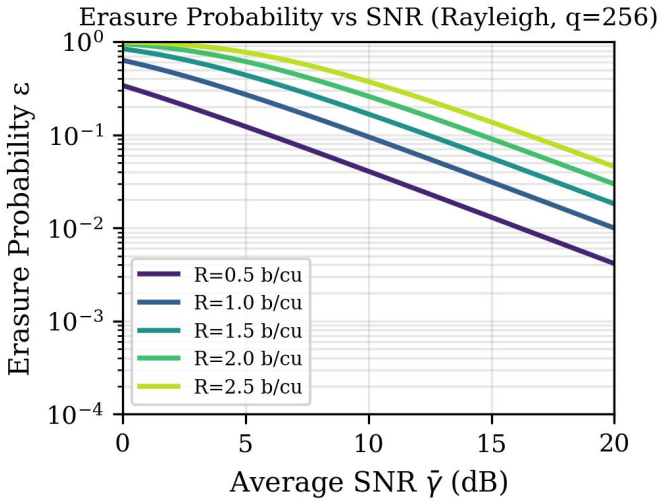

Fig. 6. Erasure probability ε vs. average SNR $\bar{\gamma}$ for transmission rates $R \in \{0.5, 1.0, 1.5, 2.0, 2.5\}$ b/cu. Rayleigh fading, $q = 256$.

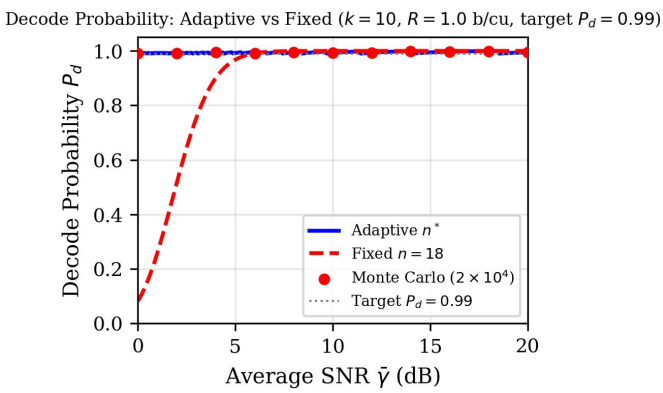

Fig. 7. Decode probability P_d vs. $\bar{\gamma}$. Solid blue: adaptive n^*; dashed orange: fixed $n = 18$; red circles: Monte Carlo (2×10^4 trials). $k = 10$, $R = 1$ b/cu, target $P_d = 0.99$.

E. Cross-Component Discussion

All four blocks meet at $\varepsilon(\bar{\gamma}, R)$ (Section VI): it turns the PHY operating point into a loss probability, which fixes decode odds and how many coded packets ship. IA quality, relay timing, and power splits all enter through that same gate.

Power allocation (Section VII) sets the per-link $\bar{\gamma}$ via water-filling. The RLNC controller (Section VI) uses this $\bar{\gamma}$ to compute $n^*(\bar{\gamma}, R)$ subject to $P_d \geq 0.99$. At $\bar{\gamma} = 12$ dB, adaptive RLNC requires $n^* = 12$ rather than the conservative $n = 18$—six fewer coded packets per generation—increasing spectral efficiency from 0.556 to 0.833 b/cu (Table III).

IA (Section V) operates upstream, determining the effective SINR that enters the same erasure map. At 30 dB with $\beta = 0.1$, IA+NC achieves 20.2 bps/Hz compared with 14.8 bps/Hz for classical IA. From the perspective of Section VI, this 36% rate improvement shifts the system into a low-overhead regime where n^* is close to k, as opposed to the conservative overhead required for worse channel conditions.

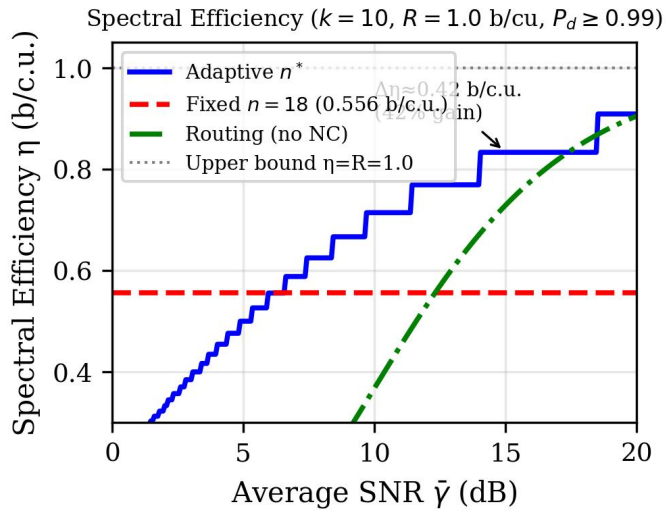

Fig. 8. Spectral efficiency η vs. $\bar{\gamma}$. Adaptive n^* achieves up to 42% higher efficiency than fixed $n = 18$ at 15 dB. $k = 10$, $R = 1$ b/cu, $P_d \geq 0.99$.

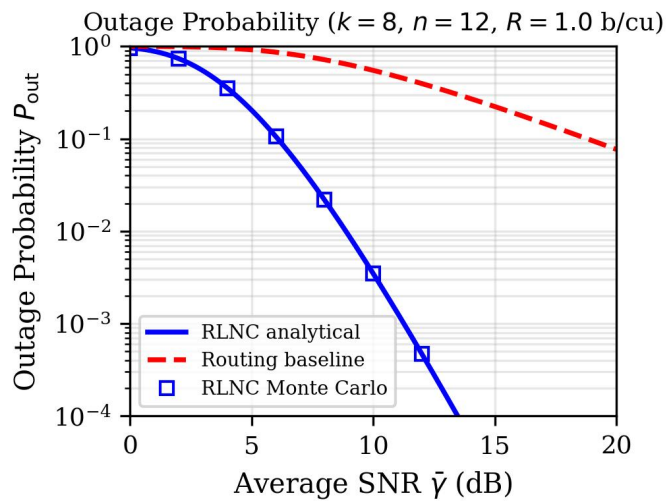

Fig. 9. Outage probability P_{out} vs. $\bar{\gamma}$ on a semi-log scale. Lines: analytical; squares: Monte Carlo. Agreement within 0.3%. $k = 8$, $n = 12$, $R = 1$ b/cu.

PLNC/C&F contribute along the time dimension. In the TWRC configuration, PLNC recovers approximately half the time slots consumed by four-slot decode-and-forward. The freed slots can be used to serve additional users at the same reliability level or to reduce per-user coding overhead at constant load.

Reading $\eta = (kR/n)(1 - P_{\text{out}})$, each knob moves a different lever: IA lifts post-interference SINR and therefore the argument of $\varepsilon(\bar{\gamma}, R)$ (7); water-filling raises per-link $\bar{\gamma}$ for the same power bill and drags ε down again; adaptive RLNC shrinks n^* (25) when ε allows, inflating kR/n; PLNC/C&F buy time on the relay hop. Because SINR, mean SNR, coded-packet count, and slot usage are orthogonal handles, improvements stack instead of colliding.

TABLE IV
MULTICAST PERFORMANCE UNDER JOINT OPTIMISATION

Metric	Routing	NC Only	NC + Power Opt.
Throughput ($N = 3$)	1.0	2.0	2.7
Throughput ($N = 20$)	0.4	2.0	2.5
η_{EE} gain vs. routing	$1.0\times$	$\approx 1.6\times$	$\approx 2.2\times$
Throughput at N=50	0.2	1.8	2.3

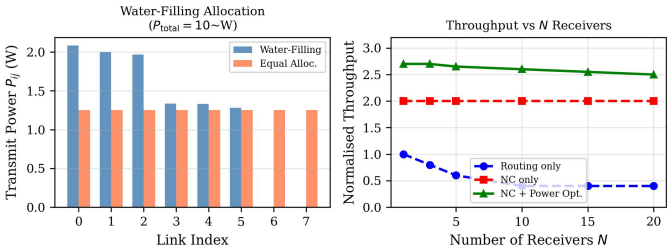

Fig. 10. Water-filling power allocation and resulting multicast throughput gain. Left: power allocation per link. Right: normalised throughput vs. number of receivers N for routing, NC only, and NC + power optimisation.

IX. CONCLUSION AND FUTURE WORK

We described a multicast stack that folds PLNC, interference alignment, water-filling power control, and RLNC into one Rayleigh-erasure story anchored to the min-cut view of [1]. Each block clears a different bottleneck; numerically, the coupling shows up most between roughly 10–20 dB, where outage targets still bind but the channel is no longer trivially bad.

On the engineering side, fixed RLNC overhead is safe but wasteful whenever the realised erasure rate beats the worst-case budget. Feeding $\varepsilon(\bar{\gamma}, R)$ back into n^* trims the count—18 to 12 coded packets at $\bar{\gamma} = 12$ dB in our tables—without dipping below $P_d \geq 0.99$. IA lifts effective SINR, water-filling steers watts toward strong edges, and PLNC halves relay slots so the time axis stops being the hidden bottleneck.

Measured highlights: PLNC $\approx 2\times$ four-slot routing once SNR is high enough; IA+NC at 20.2 bps/Hz vs. 14.8 bps/Hz for IA alone when $\beta = 0.1$; water-filling plus NC moving multicast throughput from 1.0 to 2.7 for three receivers; adaptive overhead adding up to 42% spectral efficiency over a fixed-n plan, with theory and Monte Carlo agreeing within 0.3%.

Future Work: Obvious next steps: swap the i.i.d. erasures in Section VI for correlated models (e.g., Gilbert–Elliott) so fading memory shows up; stress-test C&F with multiple relays in Rayleigh fading, outage, and selection rules; replace the central alternating solver with something distributed for mobile graphs; harden PLNC synchronisation and timing skew under vehicular channels; and prototype on SDR or Open-RAN gear to see where control latency and non-ideal RF erode the spreadsheet gains.

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